

GENETICS 101 · RECITATION HOMEWORK

Homework 2: Newborn Screening

Name: _____

Date: _____

Instructions: About 20 minutes. Part A is the chain – get that and the rest follows. Part C is a real-style batch of screening results to read like the doctor who gets the phone call. Total: **24 points**.

Part A — The chain

1. Fill in the chain for PKU, one link per row. (4 pts)

Link	For PKU, this is...
Gene (the recipe)	
Protein / enzyme (the worker)	
Job (what the worker does)	
If the recipe has a typo...	

Part B — Decode the recipe

A pretend code, three DNA letters per word: ATG = P, CCA = K, GGT = U, TTA = D, GAC = I, CTG = E, AAT = T. A word not in the table means the cell can't read it – a broken recipe.

2. (a) Decode: ATG CCA GGT. (b) Decode: TTA GAC CTG AAT – what does that recipe spell, and why is it the cure? (c) Now one letter is changed: ATG CCA GCT. What happens when the cell tries to read it? (3 pts)

Part C — Read the results

3. Four babies' newborn screens came back. For each, write what the result means and what should happen next. Remember: a positive SCREEN means 'test again, properly, now' — not a diagnosis. (1 pt per row for meaning + next step; 2 pts for the last question.) (6 pts)

Baby	Result	What it means	Next step
A	All normal; hearing pass; pulse ox pass		
B	Phenylalanine HIGH		
C	Thyroid hormone LOW		
D	Pulse ox: right hand 97%, foot 89%		

4. Which of the four babies needs action FIRST, and why? (2 pts)

Part D — Sort the conditions

5. Draw a line from each condition to what's wrong. (4 pts)

PKU	One wrong letter in the hemoglobin recipe; red cells bend
Congenital hypothyroidism	An enzyme that can't break down milk sugar
Sickle cell disease	The thyroid makes too little of its hormone
Galactosemia	A broken enzyme lets phenylalanine pile up

Part E — Flashbacks

6. From NICU Tutor: the pulse ox is used as a heart screen at 24 hours. Where are the two probes placed, and what is the machine looking for? (1 pt)

7. From last week's Genetics: cells read DNA three letters at a time. How many different three-letter words are possible? (1 pt)

Part F — Think like a doctor

8. A family gets a phone call: their baby's heel-prick screen came back positive for PKU. The father asks, "So our baby has PKU?" Answer him honestly, using what you learned about screens as nets — and say what you'd do this week. (3 pts)

HOME LAB (optional, not graded): A working enzyme and a broken one

You need: a raw potato, a cooked piece of the same potato (boiled or microwaved, then cooled), hydrogen peroxide from the first-aid kit, two small cups, a spoon, a grown-up

1. Put a slice of RAW potato in one cup and a slice of COOKED potato in the other.
2. Pour a spoonful of hydrogen peroxide over each slice. Watch for a minute.
3. Which cup fizzes? Feel the fizzing cup — is it warm? What do you think the bubbles are?
4. Write down what each cup did, and which one is 'PKU'.

What I saw:

Done! Professor's initials: _____

Score: _____ / 24 points

Professor's comment:
